Lecture 4 Scribe

L4_A.pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 4 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is the **lecture scribe for Lecture 4 — Joint Probability and Conditional Probability**, written **only from the provided slides** and organized for **closed-notes exam preparation**, with **explicit reasoning included inside the scribe**, exactly following the lecture flow.
All content is derived from the uploaded slides.

Lecture 4 Scribe — Joint Probability and Conditional Probability

1. Introduction to Probability Theory

1.1 Experiment, Outcome, Event, Sample Space

Definition: Experiment (E)
An experiment is a procedure we perform that produces some result.

Example:
Tossing a coin five times → experiment E_5

Reasoning:
We need a formal way to describe any random process.
Calling it an experiment allows us to define outcomes and probabilities precisely.

Definition: Outcome (ξ)
An outcome is a possible result of an experiment.

Example:
One outcome of tossing 5 coins:

$$\xi_1 = HHTHT$$

Reasoning:
Probability is assigned to outcomes, so we must list all possible results first.

Definition: Event
An event is a set of outcomes.

Example:

Event C = all outcomes with even number of heads.

Reasoning:
We are usually interested in conditions, not single outcomes.
So events are defined as subsets of outcomes.

Definition: Sample Space (S)
Sample space = set of all possible outcomes.

Properties:

- Mutually exclusive → cannot occur together
- Collectively exhaustive → cover all possibilities

Example: Coin flip

$$S = \{H, T\}$$

Reasoning:
Probability must sum to 1 over all possible outcomes, so we need the full set.

Types of Sample Space

- Discrete
- Countably infinite
- Continuous

Examples:

Experiment	Type
Flip coin	Discrete
Roll dice	Discrete
Flip until tails	Countably infinite
Random number in [0,1]	Continuous

Reasoning:
Different probability rules apply depending on type of sample space.

---

2. Axioms of Probability

We now assign probability to events.

Probability = numerical measure of likelihood.

Axioms (no proof needed):

1. $0 \leqslant P(A) \leqslant 1$
2. $P(S) = 1$
3. If $A \cap B = \varnothing$

$$P(A \cup B) = P(A) + P(B)$$

For infinite sets:

$$P\left(\bigcup A_i\right) = \sum P(A_i)$$

Reasoning:

- Probability cannot be negative
- Total probability must be 1
- Non-overlapping events add

These are fundamental rules used everywhere.

---

3. Corollary from Axioms

For finite mutually exclusive events:

$$P\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} P(A_i)$$

Reasoning:

Axiom 3 works for infinite sets,
so it must also work for finite sets.

---

4. Propositions from Axioms

Proposition 2.1

$$P(A^c) = 1 - P(A)$$

Reason:

A and A^c cover all outcomes.

Proposition 2.2

If $A \subset B$

$$P(A) \leqslant P(B)$$

Reason:

B contains all outcomes of A plus more.

Proposition 2.3

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Reason:

Intersection counted twice → subtract once.

Proposition 2.4 (Inclusion-Exclusion)

General formula for many events.

Reason:

Avoid double counting multiple overlaps.

---

5. Assigning Probability

Classical approach

All outcomes equally likely.

Example 1: Coin

$$P(H) = 1/2$$

Example 2: Dice

$$P(even) = 3/6$$

Example 3: Sum = 5 in two dice

Outcomes:

(1,4),(2,3),(3,2),(4,1)

$$P = 4/36$$

Reason:

Count favorable / total.

Relative frequency approach

Repeat experiment many times.

$$P(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Reason:

Probability = long-run frequency.

Limitation:

Need infinite trials.

---

6. Joint Probability

Motivation

Events may occur together.

We want:

$$P(A, B) = P(A \cap B)$$

Reason:

Not all events are mutually exclusive.

Notation

$$P(A, B)$$

or

$$P(A \cap B)$$

Multiple events:

$$P(A_1, A_2, ..., A_M)$$

Calculation

Classical:

Find common outcomes.

Relative frequency:

$$P(A, B) = \lim \frac{n_{AB}}{n}$$

Reason:

Joint probability = both occur.

Example 1: Card deck

52 cards

A = red
B = number card
C = heart

Individual:

$$P(A) = 26/52$$

$$P(B) = 40/52$$

$$P(C) = 13/52$$

Joint:

$$P(A, B) = 20/52$$

$$P(A, C) = 13/52$$

$$P(B, C) = 10/52$$

Reason:

Count common cards.

Example 2: Costume problem

Tops = 4
Bottoms = 6

Cape = 1/4
Boxers = 4/6

Joint:

$$P = 1/4 \times 4/6 = 1/6$$

Reason:

Independent choices → multiply.

---

7. Conditional Probability

Motivation

Events may depend on each other.

We want probability of A given B.

Definition

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

where $P(B) > 0$

Reason:

Restrict sample space to B.

Product Rule

$$P(A, B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

Reason:

Rewrite definition.

Three events

$$P(A, B, C) = P(C \mid A, B)P(B \mid A)P(A)$$

Chain rule for many events.

Reason:

Break joint probability step-by-step.

---

8. Example — Cards without replacement

A = first card spade
B = second card spade

After A:

Total = 51
Spades = 12

$$P(B \mid A) = 12/51$$

Reason:

Sample space changed.

---

9. Example — Poker flush

Flush in spades:

$$13/52 \times 12/51 \times 11/50 \times 10/49 \times 9/48$$

Any flush:

4 suits

$$4 \times P(spadeflush)$$

Reason:

Mutually exclusive suits.

---

10. Example — Missing key

P(L)=0.4

P(R)=0.4

P(K)=0.8

Find:

$$P(R \mid L^c) = \frac{P(R)}{1 - P(L)} = 0.4/0.6 = 2/3$$

Reason:

Right pocket implies not left.

Final Summary (Exam Focus)

You must remember:

- Definitions: experiment, outcome, event, sample space
- Axioms of probability
- Complement rule
- Union rule
- Joint probability
- Conditional probability formula
- Product rule
- Chain rule
- Classical vs relative frequency
- Card / dice / poker / key examples

These allow solving any exam problem.

✅ Scribe finished
✅ Faithful to slides
✅ Step-by-step reasoning included
✅ Exam-ready format

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Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is a **faithful lecture scribe for Lecture 4 — Joint Probability and Conditional Probability**, written **only from the provided slides**, with **step-by-step reasoning**

included inside the scribe, and organized for **closed-notes exam preparation**.
Content is taken from the uploaded Lecture-4 slides.

Lecture 4 Scribe

Joint Probability and Conditional Probability

---

1. Introduction to Probability Theory

1.1 Experiments, Sample Space, and Events

Definition: Experiment (E)

An experiment is a procedure we perform that produces some result.

Example:
Tossing a coin five times → experiment E_5

Reasoning:
Probability theory studies random processes.
To analyze randomness mathematically, we first define the process as an experiment.

Definition: Outcome (ξ)

An outcome is a possible result of an experiment.

Example:

One outcome of E_5:

$$\xi_1 = HHTHT$$

Reasoning:
Probability is assigned to outcomes, so we must identify all possible results of the experiment.

Definition: Event

An event is a set of outcomes of an experiment.

Example:

Event C for experiment E_5:

C = all outcomes with even number of heads

Reasoning:
We are usually interested in conditions (like even heads), not single outcomes.
So events are defined as sets of outcomes.

Definition: Sample Space (S)

Sample space is the set of all possible outcomes.

Properties:

- Mutually exclusive → cannot occur together
- Collectively exhaustive → cover all possibilities

Example:

Flip coin once:

$$S = \{H, T\}$$

Reasoning:
To assign probability correctly, we must list all possible outcomes.

Sample Space (continued)

Sample space is the universal set of outcomes.

Types:

- Discrete
- Countably infinite
- Continuous

Examples:

- Flip coin once → discrete
- Roll dice → discrete
- Flip until tails → infinite
- Random number in [0,1] → continuous

Reasoning:
Different types of sample spaces require different probability models.

---

2. Axioms of Probability

Next step: assign probability to events.

Probability:

A numerical measure of likelihood.

Axioms (self-evident rules):

1.

$$0 \leqslant Pr(A) \leqslant 1$$

Reason: probability cannot be negative or greater than 1.

2.

If S is the sample space,

$$Pr(S) = 1$$

Reason: one of the outcomes must occur.

3.

If $A \cap B = \varnothing$,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

Reason: no overlap, so probabilities add.

For infinite mutually exclusive events:

$$Pr(\bigcup A_i) = \sum Pr(A_i)$$

Reason: total probability equals sum of individual probabilities.

---

3. Corollary from Probability Axioms

For finite mutually exclusive events:

$$Pr(\bigcup_{i=1}^{M} A_i) = \sum_{i=1}^{M} Pr(A_i)$$

Reasoning:

Axiom works for infinite sets → must also work for finite sets.

When sample space is finite, this corollary is equivalent to axiom.

---

4. Propositions from Probability Axioms

Proposition 2.1

$$Pr(A^c) = 1 - Pr(A)$$

Reason:

A and complement cover entire sample space.

Proposition 2.2

If $A \subset B$,

$$Pr(A) \leqslant Pr(B)$$

Reason:

B contains all outcomes of A.

Proposition 2.3

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Reason:

Intersection counted twice → subtract once.

Proposition 2.4 (General form)

For many events:

Use inclusion-exclusion formula.

Reason:

Avoid multiple counting of overlaps.

---

5. Assigning Probabilities

5.1 Classical Approach

Assumption:

All outcomes equally likely.

Example 1:

Coin flip

$$Pr(H) = 1/2$$

Reason:

Two equally likely outcomes.

Example 2:

Dice roll

Even numbers:

2,4,6

$$Pr(even) = 3/6$$

Reason:

Count favorable / total.

Example 3:

Two dice, sum = 5

Outcomes:

(1,4), (2,3), (3,2), (4,1)

Total = 36

$$Pr = 4/36$$

Reason:

Each pair equally likely.

5.2 Relative Frequency Approach

Repeat experiment many times.

$$Pr(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Reason:

Probability equals long-run frequency.

Limitation:

Need infinite trials.

Some experiments not repeatable.

---

6. Joint Probability

Motivation

Not all events are mutually exclusive.

We want probability of both events.

$$Pr(A, B) = Pr(A \cap B)$$

Reason:

Events may occur together.

Notation

$$Pr(A, B)$$

or

$$Pr(A \cap B)$$

Multiple events:

$$Pr(A_1, A_2, \dots, A_M)$$

Reason:

Need notation for simultaneous events.

Calculation

Classical method

- Express A and B using outcomes
- Find common outcomes
- Count probabilities

Reason:

Joint event = intersection.

Relative frequency

Let n_{AB} = times both occur.

$$Pr(A, B) = \lim \frac{n_{AB}}{n}$$

Reason:

Joint probability = frequency of both.

Example 1 — Card Deck

52 cards

A = red
B = number card
C = heart

Individual:

$$Pr(A) = 26/52$$

$$Pr(B) = 40/52$$

$$Pr(C) = 13/52$$

Joint:

$$Pr(A, B) = 20/52$$

$$Pr(A, C) = 13/52$$

$$Pr(B, C) = 10/52$$

Reason:

Count cards satisfying both.

Example 2 — Costume Party

Tops:

3 shirts + 1 cape = 4

$$Pr(cape) = 1/4$$

Bottoms:

2 pants + 4 boxers = 6

$$Pr(boxers) = 4/6$$

Joint:

$$Pr = 1/4 \times 4/6 = 1/6$$

Reason:

Independent selections.

Multiply probabilities.

---

7. Conditional Probability

Motivation

One event may depend on another.

Example:

Second card depends on first.

Definition

$$Pr(A \mid B) = \frac{Pr(A, B)}{Pr(B)}$$

where $Pr(B) > 0$

Reason:

Restrict sample space to B.

Product Rule

$$Pr(A, B) = Pr(A \mid B)Pr(B) = Pr(B \mid A)Pr(A)$$

Reason:

Rewrite definition.

Three events

$$Pr(A, B, C) = Pr(C \mid A, B)Pr(B \mid A)Pr(A)$$

Reason:

Break joint probability step by step.

Chain rule

For many events:

Multiply conditional probabilities.

Reason:

Each step depends on previous.

---

8. Example — Cards without replacement

A = first card spade
B = second card spade

Initial:

52 cards
13 spades

After A:

51 cards
12 spades

$$Pr(B \mid A) = 12/51$$

Reason:

Sample space changes.

---

9. Example — Poker Flush

Flush = 5 cards same suit

Spade flush:

$$13/52 \times 12/51 \times 11/50 \times 10/49 \times 9/48$$

Reason:

Without replacement.

Any flush:

4 suits

$$4 \times P(spadeflush)$$

Reason:

Mutually exclusive suits.

---

10. Example — Missing Key

Given:

P(L)=0.4
P(R)=0.4
P(K)=0.8

Find:

$$P(R \mid L^c) = \frac{P(R)}{1 - P(L)} = 0.4/0.6 = 2/3$$

Reason:

Right pocket implies not left.

Final Exam Notes

Remember:

- Definitions
- Axioms
- Complement rule
- Union rule
- Classical probability
- Relative frequency
- Joint probability
- Conditional probability
- Product rule
- Chain rule
- Card example
- Dice example
- Costume example
- Poker example
- Missing key example

These are exactly the concepts used in exam problems.

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Below is a **fully faithful, step-by-step lecture scribe for Lecture 4 — Joint Probability and Conditional Probability**, written **only from the provided slides**, with **explicit reasoning included inside the scribe**, and structured exactly in lecture order so a student can **reconstruct definitions, proofs, and examples in a closed-notes exam**.
All statements come from the uploaded Lecture-4 slides.

Lecture 4 Scribe

Joint Probability and Conditional Probability

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1. Introduction to Probability Theory

1.1 Experiments, Sample Space, and Events

Definition: Experiment (E)

An experiment is a procedure we perform that produces some result.

Example:
Tossing a coin five times $\rightarrow E_5$

Reasoning:
Probability theory studies uncertain processes.

To analyze uncertainty mathematically, the first step is to define the process as an experiment.

Definition: Outcome (ξ)

An outcome is a possible result of an experiment.

Example:

One outcome of E_5:

$$\xi_1 = HHTHT$$

Reasoning:
Probability will later be assigned to outcomes, so we must list all possible results of the experiment.

Definition: Event

An event is a set of outcomes of an experiment.

Example:

Event C in experiment E_5

C = all outcomes with even number of heads

Reasoning:
We are usually interested in conditions (like even heads), not a single outcome. Therefore events are defined as subsets of the sample space.

Definition: Sample Space (S)

Sample space is the collection of all possible outcomes.

Properties:

- Mutually exclusive → only one outcome occurs
- Collectively exhaustive → all possibilities included

Example:

Flip coin once

$$S = \{H, T\}$$

Reasoning:
To assign probabilities, we must know every possible outcome.

Sample Space (continued)

Sample space is the universal set of outcomes.

Types:

- Discrete
- Countably infinite
- Continuous

Examples:

Experiment	Type
Flip coin	Discrete
Roll die	Discrete
Flip until tails	Countably infinite
Random number in [0,1]	Continuous

Reasoning:
Different sample spaces require different probability models.

---

2. Axioms of Probability

Next step: assign probability to events.

Probability:

A numerical measure of likelihood.

Axioms (self-evident rules):

Axiom 1

$$0 \leqslant Pr(A) \leqslant 1$$

Reasoning:

Probability cannot be negative
Probability cannot exceed certainty.

Axiom 2

If S is the sample space,

$$Pr(S) = 1$$

Reasoning:

One of the outcomes must occur.

Axiom 3

If $A \cap B = \varnothing$,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

Reasoning:

Events do not overlap, so probabilities add.

For infinite mutually exclusive events:

$$Pr\left(\bigcup A_i\right) = \sum Pr(A_i)$$

Reasoning:

Total probability equals sum of each disjoint event.

---

3. Corollary from Probability Axioms

For finite mutually exclusive events:

$$Pr(\bigcup_{i=1}^{M} A_i) = \sum_{i=1}^{M} Pr(A_i)$$

Reasoning:

Axiom works for infinite sets
→ must also work for finite sets.

When sample space is finite, this is equivalent to axiom.

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4. Propositions from Probability Axioms

Proposition 2.1

$$Pr(A^c) = 1 - Pr(A)$$

Reasoning:

A and complement together form the whole sample space.

Proposition 2.2

If $A \subset B$,

$$Pr(A) \leqslant Pr(B)$$

Reasoning:

B contains all outcomes of A and possibly more.

Proposition 2.3

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Reasoning:

Intersection counted twice → subtract once.

Proposition 2.4 (General union rule)

For many events:

Use inclusion-exclusion formula.

Reasoning:

Avoid counting overlaps multiple times.

---

5. Assigning Probabilities

5.1 Classical Approach

Assumption:

All outcomes equally likely.

Example 1 — Coin

$$Pr(H) = 1/2$$

Reason:

2 equally likely outcomes.

Example 2 — Dice

Even numbers = {2,4,6}

$$Pr(even) = 3/6$$

Reason:

Count favorable / total.

Example 3 — Two dice sum = 5

Outcomes:

(1,4), (2,3), (3,2), (4,1)

Total outcomes = 36

$$Pr = 4/36$$

Reason:

Each pair equally likely.

5.2 Relative Frequency Approach

Repeat experiment many times.

$$Pr(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Reasoning:

Probability = long-run frequency.

Limitation:

Need infinite trials.
Some events cannot be repeated.

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6. Joint Probability

Motivation

Not all events are mutually exclusive.

We want probability of both events.

$$Pr(A, B) = Pr(A \cap B)$$

Reasoning:

Events may occur together.

Notation

$$Pr(A, B)$$

or

$$Pr(A \cap B)$$

Multiple events:

$$Pr(A_1, A_2, \dots, A_M)$$

Reason:

Need notation for simultaneous events.

Calculation Methods

Classical

1. Express A and B using outcomes
2. Find common outcomes
3. Count probabilities

Reason:

Joint event = intersection.

Relative frequency

Let n_{AB} = times both occur.

$$Pr(A, B) = \lim \frac{n_{AB}}{n}$$

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Joint probability = frequency of both.

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Reasoning:

Count cards satisfying both conditions.

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Tops:

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Bottoms:

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$$Pr(boxers) = 4/6$$

Joint probability:

$$Pr = 1/4 \times 4/6 = 1/6$$

Reasoning:

Selections independent
→ multiply probabilities.

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7. Conditional Probability

Motivation

One event may depend on another.

Example:

Second card depends on first.

Definition

$$Pr(A \mid B) = \frac{Pr(A, B)}{Pr(B)}$$

where $Pr(B) > 0$

Reasoning:

When B occurs, sample space becomes B.

Product Rule

$$Pr(A, B) = Pr(A \mid B)Pr(B) = Pr(B \mid A)Pr(A)$$

Reason:

Rewrite definition.

Three events

$$Pr(A, B, C) = Pr(C \mid A, B)Pr(B \mid A)Pr(A)$$

Reason:

Break joint probability step-by-step.

Chain Rule

For many events:

Multiply conditional probabilities.

Reason:

Each step depends on previous.

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8. Example — Cards without replacement

A = first card spade
B = second card spade

Initial:

52 cards
13 spades

After A:

51 cards
12 spades

$$Pr(B \mid A) = 12/51$$

Reason:

Sample space changed after first draw.

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9. Example — Poker Flush

Flush = 5 cards same suit

Spade flush:

$$13/52 \times 12/51 \times 11/50 \times 10/49 \times 9/48$$

Reason:

Cards drawn without replacement.

Any flush:

4 suits

$$4 \times P(spadeflush)$$

Reason:

Suits mutually exclusive.

---

10. Example — Missing Key

Given:

P(L)=0.4
P(R)=0.4
P(K)=0.8

Find:

$$P(R \mid L^c) = \frac{P(R)}{1 - P(L)} = 0.4/0.6 = 2/3$$

Reason:

Right pocket implies not left pocket.

Final Exam Summary

You must remember:

- Experiment, outcome, event, sample space
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- Complement rule
- Union rule
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- Product rule
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Is this conversation helpful so far?